Life Cycle of Data Science in Indian Stock Market

1. Business Understanding

Meaning

Business understanding is the first step where we identify the problem and goal of the stock market project.
In the Indian stock market, the goal may be predicting stock prices, reducing investment risk, or finding profitable stocks.This step helps to understand what the company or investor needs.
It also defines success criteria and business objectives clearly.

Example

Predict whether Reliance Industries stock price will increase or decrease in the next week.

2. Data Collection

Meaning

Data collection means gathering stock market data from different sources.
The data may include stock prices, trading volume, company financial reports, and news data.Accurate and large amounts of data improve prediction quality.Data can be collected from APIs, websites, or stock exchanges.

Example

Collect daily stock prices of Tata Consultancy Services from NSE India.

3. Data Preprocessing

Meaning

Data preprocessing is the process of cleaning and preparing data for analysis.It removes missing values, duplicate records, and incorrect data.The data is converted into a proper format for machine learning models.This step improves data quality and accuracy.

Example

Remove missing stock price values and convert date format in Infosys stock dataset.

4. Exploratory Data Analysis (EDA)

Meaning

EDA is used to understand patterns, trends, and relationships in stock market data.Graphs and charts help identify market behavior and stock movement.It helps to find outliers, volatility, and correlations between variables.EDA gives better insights before building models.

Example

Analyze how Infosys stock price changes during market crashes.

5. Model Building

Meaning

Model building is the step where machine learning algorithms are trained using stock data.
The model learns patterns from historical market data.Different algorithms like Linear Regression, Random Forest, or LSTM can be used.The goal is to predict future stock prices accurately.

Example

Use Linear Regression to predict the next day closing price of HDFC Bank stock.

6. Testing and Evaluation

Meaning

Testing and evaluation check how well the model performs on new data.Metrics like Accuracy, RMSE, and MAE are used to measure performance.This step helps to know whether the model predictions are reliable.A good model should provide accurate and consistent results.

Example

Compare predicted and actual stock prices of HDFC Bank.

7. Deployment

Meaning

Deployment means making the trained model available for real-world use.The model can be integrated into websites, mobile apps, or trading systems.Users can access stock predictions directly through applications.This step converts the model into a usable business solution.

Example

Deploy a stock prediction system in a trading dashboard for investors.

8. Monitoring

Meaning

Monitoring is the continuous checking of model performance after deployment.Stock market trends change frequently, so the model accuracy may reduce over time.The model should be updated with new market data regularly.Monitoring helps maintain prediction accuracy and system reliability.

Example

Retrain the model monthly using latest Indian stock market data.